

SEGMENT 1

Concepts of System and Information

1. System

Definition

A **system** is a collection of interrelated components that work together to achieve a common objective.

A system transforms **input into output**.

General model of a system:

Input → Processing → Output
 ↑
 Feedback

Example:

Banking system

Input: customer transaction
Processing: verify and update balance
Output: updated account information

2. Characteristics of a System

A system has several important characteristics.

1 Organization

A system has a structured arrangement of components.

Example

Hospital system includes

patients
doctors
nurses
billing
records

2 Interaction

Components interact with each other.

Example

Bank system

ATM interacts with database

3 Interdependence

Subsystems depend on each other.

Example

sales system depends on inventory system

4 Integration

All components work together toward a common objective.

Example

Online shopping system integrates

payment
inventory
shipping

5 Goal Oriented

Every system works toward a goal.

Example

University system goal:

education and student management

3. System Environment

The **environment of a system** includes everything outside the system that affects it.

Example

For a company system:

Environment includes

customers
competitors
government
suppliers
technology

4. Components of a System

Every system consists of several components.

1 Input

Data entering the system.

Example

customer order

2 Process

Operations performed on data.

Example

`calculate total price`

3 Output

Results produced by the system.

Example

`invoice`

4 Feedback

Information returned to improve the system.

Example

`customer review`

5 Control

Mechanisms used to regulate the system.

Example

`access control`

5. Types of Systems

Systems can be classified in different ways.

5.1 Physical vs Abstract System

Physical System

A system that exists physically.

Examples

```
computer system  
transport system  
manufacturing system
```

Abstract System

Conceptual system based on ideas or models.

Examples

```
mathematical model  
economic model
```

5.2 Open vs Closed System

Open System

Interacts with the environment.

Example

```
business organization
```

Closed System

Does not interact with the environment.

Example

a completely self-contained machine

5.3 Deterministic vs Probabilistic System

Deterministic System

Behavior is predictable.

Example

mathematical calculation system

Probabilistic System

Behavior involves uncertainty.

Example

weather forecasting
stock market

6. Information

Definition

Information is **processed data that is meaningful and useful for decision making.**

Example

Raw data

100
120
150

Processed information

sales increased by 25%

7. Data vs Information

Data	Information
raw facts	processed data
unorganized	meaningful
no context	useful for decisions

Example

Data

45, 60, 55

Information

average marks = 53

8. Types of Information

Your exam papers often ask:

Strategic, Tactical and Operational information

8.1 Strategic Information

Used by **top-level management**.

Characteristics

long-term decisions
future planning

Examples

market expansion
new product development

Example from your paper

Foodpanda planning new feature strategy

8.2 Tactical Information

Used by **middle management**.

Characteristics

medium-term decisions
department planning

Examples

sales targets
marketing campaign planning

8.3 Operational Information

Used by **lower management**.

Characteristics

day-to-day operations
short-term decisions

Examples

daily sales report
inventory update

9. Quality / Characteristics of Information

Your papers repeatedly ask this question.

Important characteristics include:

Accuracy

Information must be correct.

Example

correct financial report

Timeliness

Information must be available at the right time.

Example

real-time stock price

Relevance

Information must be useful for the decision.

Example

customer demand data for marketing team

Completeness

Information must not be missing important details.

Example

complete sales report

Reliability

Information must come from trusted sources.

Example

verified financial data

Consistency

Information should not contradict other data.

Accessibility

Information must be easily accessible.

Example

dashboard reports

10. Sources of Information

Information can come from **internal** or **external** sources.

10.1 Internal Sources

Generated inside the organization.

Examples

sales records
financial reports
employee records
inventory data

10.2 External Sources

Come from outside the organization.

Examples

government reports
market research
competitors
customers
suppliers

11. Example From Previous Question (Foodpanda)

Example statements:

Pop-up notifications

Type:

Operational information

Reason

supports daily user interaction

Top 10 ordered meals

Type:

Tactical information

Reason

helps marketing and product decisions

Calorie count feature

Type:

Strategic information

Reason

long-term product improvement

SEGMENT 2

Information Gathering & System Development Methodologies

1. Information Gathering

Information gathering is the process of **collecting facts and requirements about the existing system and the proposed system.**

It is one of the most important tasks of a **system analyst.**

Purpose of information gathering:

- understand user requirements
- identify system problems
- analyze current processes
- determine system improvements

Example

Before developing an **e-commerce website**, the analyst gathers information about:

customer needs
payment methods
inventory system
delivery process

2. Information Gathering Strategy

A strategy defines **how the analyst collects information efficiently**.

Common strategies include:

1 Top-Down Strategy

Start from **overall system understanding**, then move to details.

Example

organization → departments → processes → tasks

2 Bottom-Up Strategy

Start from **detailed tasks**, then understand the whole system.

Example

employee tasks → department processes → organizational system

3 Middle-Out Strategy

Start from the **most important processes**, then expand upward and downward.

Example

focus on payment system first → then connect inventory and delivery

3. Information Searching Methods

System analysts use several methods to collect information.

The most common methods include:

3.1 Interview

Interview is a **direct conversation between the analyst and system users.**

Purpose:

- gather detailed information
- understand user problems
- identify requirements

Types of interviews

Structured Interview

Questions are **predefined.**

Example

What problems do you face while using the system?

Unstructured Interview

Questions are **open and flexible.**

Example

Explain how the system works in your department.

3.2 Questionnaire

A questionnaire is a **written set of questions distributed to many users.**

Advantages

collect information from many people quickly
low cost

Disadvantages

limited detailed explanation

3.3 Observation

The analyst **observes how the system actually works.**

Example

Watching how employees process customer orders.

Advantages

real understanding of system operations

Disadvantages

time consuming

3.4 Document Review

The analyst studies **existing documents.**

Examples

reports
forms
manuals
database records

3.5 Sampling

Instead of analyzing all data, the analyst studies **a sample of data.**

Example

review 100 invoices instead of 10,000 invoices

4. Interviewing Technique

A good interview requires proper planning.

Steps of interviewing:

Step 1 Preparation

The analyst prepares:

questions
objectives
background information

Step 2 Conducting Interview

During the interview:

ask clear questions
listen carefully
take notes

Step 3 Closing Interview

At the end:

summarize discussion
confirm information
thank the interviewee

Step 4 Documentation

Record the interview results for future analysis.

5. System Development Methodologies

A **methodology** is a structured approach used to develop information systems.

It defines:

process
rules
techniques
tools

Examples

Waterfall
Spiral
Agile
RAD

6. System Development Life Cycle (SDLC)

SDLC is a **structured process used to develop an information system**.

It describes the stages involved in system development.

General SDLC model:

Planning
↓
System Analysis
↓
System Design
↓
Implementation
↓
Testing
↓
Deployment
↓
Maintenance

7. Phases of SDLC

7.1 Planning Phase

This phase identifies:

project goals
project scope
resources required

Example

Planning an **online learning platform**.

7.2 System Analysis Phase

The analyst studies the **existing system** and determines requirements.

Tasks include:

requirement gathering
problem identification
feasibility study

7.3 System Design Phase

In this phase the system structure is designed.

Design includes:

database design
system architecture
user interface design

7.4 Implementation Phase

The system is **developed and coded**.

Activities

programming
database creation
integration

7.5 Testing Phase

The system is tested to ensure it works correctly.

Types of testing

unit testing
integration testing
system testing

7.6 Deployment Phase

The system is installed and delivered to users.

Example

Launching an **e-commerce website**.

7.7 Maintenance Phase

After deployment the system requires updates.

Maintenance includes

bug fixes
performance improvements
feature updates

8. SDLC Models

Different SDLC models define different ways of developing systems.

8.1 Waterfall Model

The Waterfall model follows a **linear sequence of phases**.

Structure

```
Requirements
↓
Design
↓
Implementation
↓
Testing
↓
Maintenance
```

Advantages

```
simple
easy to manage
clear structure
```

Disadvantages

```
not flexible
difficult to change requirements
```

8.2 Spiral Model

The Spiral model combines **iteration and risk analysis**.

Each cycle includes:

```
planning
risk analysis
development
evaluation
```

Advantages

good for high-risk projects
early risk detection

Disadvantages

complex
expensive

8.3 Agile Model

Agile development focuses on **iterative development and continuous feedback**.

Features

small development cycles
customer collaboration
frequent updates

Advantages

flexible
adaptable to changes
fast delivery

Disadvantages

less documentation
requires experienced team

9. Rapid Application Development (RAD)

RAD focuses on **fast system development using prototypes**.

Characteristics

rapid prototyping
user involvement
short development cycles

Advantages

faster development
high user satisfaction

Disadvantages

not suitable for very large systems

10. Joint Application Development (JAD)

JAD is a method where **users, managers, and developers collaborate in workshops** to define system requirements.

Benefits

better communication
faster decision making
accurate requirements

11. Choosing an SDLC Model

Examples:

Project with clear requirements

Best model:

Waterfall

Project with changing requirements

Best model:

Agile

Project with high risk

Best model:

Spiral

12. Methodology vs Model vs Tools vs Techniques

This appears in your exam questions.

Methodology

Overall framework for system development.

Example

Agile methodology

Model

Specific representation of development process.

Example

Waterfall model
Spiral model

Tools

Software used in development.

Examples

CASE tools
JIRA
Git

Techniques

Methods used to perform tasks.

Examples

interview
data flow diagram
prototyping

SEGMENT 3

Feasibility Study & Cost–Benefit Analysis

1. Feasibility Study

Definition

A **feasibility study** is an evaluation conducted to determine whether a proposed system or project is **practical, achievable, and beneficial** for an organization.

It helps answer the question:

“Should the project be developed or not?”

Feasibility study is usually performed during the **system analysis phase of SDLC**.

2. Importance of Feasibility Study

Feasibility study helps organizations:

- determine if the project is possible
- estimate project cost and benefits
- reduce project risks
- evaluate alternative solutions
- avoid wasting time and resources

Example

Before developing an **online banking system**, analysts check:

technical capability
cost of development
security risks
expected benefits

3. Types of Feasibility

Several types of feasibility are analyzed in system development.

3.1 Technical Feasibility

Determines whether the organization has the **technical resources and expertise** required.

Questions considered:

Do we have required hardware?
Do we have skilled developers?
Is the technology reliable?

Example

A company planning to build a **mobile banking app** must check if they have mobile development expertise.

3.2 Economic Feasibility

Determines whether the project is **financially worthwhile**.

It compares:

project costs
project benefits

If benefits exceed costs, the project is economically feasible.

Example

Developing an e-commerce platform costing \$100,000 but generating \$300,000 revenue.

3.3 Operational Feasibility

Determines whether the system will **work in the organization's environment**.

Questions:

Will employees accept the system?
Will it improve operations?
Will users be trained easily?

3.4 Schedule Feasibility

Determines whether the project can be completed **within the required time**.

Example

A system required before **holiday sales season** must be delivered on time.

3.5 Legal Feasibility

Determines whether the system follows **legal regulations**.

Example

Data privacy laws

GDPR
cyber security regulations

4. Steps in Feasibility Analysis

A typical feasibility study involves the following steps.

Step 1 Identify the Problem

Understand the system problem or opportunity.

Example

A company wants to improve its **inventory management system**.

Step 2 Identify Possible Solutions

List alternative solutions.

Example

build new system
upgrade existing system
purchase commercial software

Step 3 Evaluate Feasibility

Analyze each solution based on:

technical feasibility
economic feasibility
operational feasibility
schedule feasibility

Step 4 Select Best Alternative

Choose the solution that provides maximum benefits with minimum cost.

Step 5 Prepare Feasibility Report

Document all findings and recommendations.

5. Feasibility Report

A **feasibility report** is a document that summarizes the results of the feasibility study.

Typical contents include:

- problem description
- proposed solutions
- feasibility evaluation
- cost-benefit analysis
- recommendations

This report helps management decide **whether to approve the project**.

6. Cost–Benefit Analysis

Cost–Benefit Analysis (CBA) is a technique used to compare the **costs and benefits of a project**.

Purpose:

- determine economic feasibility
- measure project profitability
- support decision making

If:

Benefits > Costs

the project is considered worthwhile.

7. Types of Costs

Costs involved in system development can be classified into different categories.

7.1 Development Cost

Costs required to develop the system.

Examples

software development
hardware purchase
system design
testing

7.2 Operational Cost

Costs required to operate the system after deployment.

Examples

maintenance
electricity
staff salaries

7.3 Fixed Cost

Costs that remain constant.

Example

server purchase
software license

7.4 Variable Cost

Costs that change depending on usage.

Example

cloud storage cost
data transfer charges

7.5 Recurring Cost

Costs that occur repeatedly.

Example

monthly hosting fee
annual software license
staff salary

7.6 Non-Recurring Cost

One-time costs.

Example

system installation
hardware purchase

8. Types of Benefits

Benefits obtained from a system may also be classified.

8.1 Tangible Benefits

Benefits that can be **measured in money**.

Examples

increased sales
reduced labor cost
reduced operational cost

8.2 Intangible Benefits

Benefits that **cannot be easily measured financially**.

Examples

improved customer satisfaction
better decision making
enhanced company reputation

9. Cost and Benefit Determination Procedure

To perform cost–benefit analysis:

- Step 1 Identify all costs
 - Step 2 Identify all benefits
 - Step 3 Estimate cost values
 - Step 4 Estimate benefit values
 - Step 5 Compare total costs and benefits
-

10. Cost–Benefit Evaluation Methods

Several financial techniques are used.

10.1 Return on Investment (ROI)

ROI measures the **profitability of the project**.

Formula:

$$\text{ROI} = (\text{Total Benefits} - \text{Total Costs}) / \text{Total Costs} \times 100$$

Example

Total Benefits = 200,000

Total Costs = 100,000

ROI

$$\text{ROI} = (200000 - 100000) / 100000 \times 100$$

$$\text{ROI} = 100\%$$

10.2 Break-Even Point

Break-even point is the time when **total benefits equal total costs**.

After this point the project starts generating profit.

Example

If:

Total Cost = 100,000

Annual Benefit = 50,000

Break-even occurs after:

2 years

10.3 Net Present Value (NPV)

NPV evaluates project profitability by considering **time value of money**.

Formula

NPV = Present value of benefits - Present value of costs

If:

NPV > 0

the project is profitable.

10.4 Cash Flow Analysis

Cash flow shows the **movement of money over time**.

Example table from your exam.

Year	Benefits	Costs	Net Cash Flow
0	0	100000	-100000
1	45000	10000	35000
2	50000	12000	38000
3	57000	16000	41000

Cumulative cash flow is calculated to determine break-even.

11. Break-Even Example

Cumulative net cash flow:

Year 0 = -100000
Year 1 = -65000
Year 2 = -27000
Year 3 = 14000

Break-even occurs **during Year 3**, because cumulative cash flow becomes positive.

12. Deliverables

Deliverables are **tangible outputs produced during project development.**

Examples

system design documents
software modules
user manuals
test reports
